🗺️ Phase 1: Base Initialization & Manual Mapping

Do not start Nav2 yet. We need to build the map first so the robot doesn't get stuck in "unknown" space.

Terminal 1: Start Motor Driver

Bash

```
python3 ~/ros2_ws/demo/direct_driver.py
```

Terminal 2: Launch Lidar, Odometry & TF Tree *(Copy and paste this entire block)*

Bash

```
ros2 launch ldlidar_stl_ros2 ld19.launch.py & \
ros2 launch rf2o_laser_odometry rf2o_laser_odometry.launch.py & \
ros2 run tf2_ros static_transform_publisher 0 0 0 0 0 0 odom base_footprint --ros-args -p use_sim_time:=false & \
ros2 run tf2_ros static_transform_publisher 0 0 0 0 0 0 base_footprint base_link --ros-args -p use_sim_time:=false & \
ros2 run tf2_ros static_transform_publisher 0 0 0.1 0 0 0 base_link base_laser --ros-args -p use_sim_time:=false
```

Terminal 3: Start SLAM (Mapping)

Bash

```
ros2 run slam_toolbox async_slam_toolbox_node --ros-args -p use_sim_time:=false -p odom_frame:=odom -p base_frame:=base_footprint -p map_frame:=map
```

🎮 ACTION REQUIRED: Manual Exploration

- **Stop typing.** Use the manual teleop/controller to drive the rover around the lab floor slowly.
- Let the Lidar scan the environment and build a solid map in the background.

📸 Phase 2: Vision System (Red Block Detection)

Once the rover is positioned and the map is built:

Terminal 4: Start Camera & Vision Script

Bash

```
bash ~/ros2_ws/demo/start_vision.sh
```

(Wait for the camera to initialize and detect the target).

🚀 Phase 3: Autonomous Navigation

Now that the map exists and the target is found, we activate the "brain".

Terminal 5: Launch Nav2 (With Tuned Parameters) *Note: We use a custom YAML file to reduce the robot's inflation radius so it doesn't get scared of tight spaces.*

Bash

```
ros2 launch nav2_bringup navigation_launch.py use_sim_time:=False params_file:=/home/rover/ros2_ws/demo/fat_to_thin_params.yaml
```

(Wait until you see Managed nodes are active in the logs).

Terminal 6: Execute Autonomous Cruise

Bash

```
python3 ~/ros2_ws/demo/auto_cruise.py
```

🛑 Troubleshooting Quick Fix

If the system freezes or components won't launch, kill all ghost processes and clear the DDS shared memory:

Bash

```
pkill -9 -f ros2; pkill -9 -f python3; rm -rf /dev/shm/fastrtps*
```